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| **Fiscal Year:** | FY 2014 | **Task Last Updated:** | FY 03/21/2014 |
| **PI Name:** | Mudawar, Issam Ph.D. | | |
| **Project Title:** | Flow Boiling and Condensation Experiment (FBCE) | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | FLUID PHYSICS | | |
| **Program/Discipline--Element/Subdiscipline:** | FLUID PHYSICS--Fluid physics | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **City:** | West Lafayette | **State:** | IN |
| **Zip Code:** | 47907-1288 | **Congressional District:** | 4 |
| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 2008 Fluid Physics NNH08ZTT002N |
| **Start Date:** | 06/01/2012 | **End Date:** | 05/31/2017 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | 1 |
| **No. of PhD Candidates:** | 6 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA GRC |
| **Contact Monitor:** | Chao, David | **Contact Phone:** | 216-433-8320 |
| **Contact Email:** | david.f.chao@nasa.gov | | |
| **Flight Program:** | ISS | | |
| **Flight Assignment:** | ISS<br>NOTE: Extended to 5/31/2017 per NSSC information (Ed., 6/17/13)<br>NOTE: Change in grant number and to be extended, per D. Chao/GRC (Ed. 3/6/2013) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | NNX13AB01G ; NNX12AK14G | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | Several efforts are presently underway to reduce the size and weight of space systems by transitioning from single-phase to two-phase thermal management. Because of large density differences between liquid and vapor, buoyancy can play an important role in defining the motion of vapor relative to liquid and, therefore, influence heat transfer effectiveness. This is particularly the case for future space missions, including the manned mission to Mars. The emphasis in two-phase fluid physics is evident from several recent NASA workshops that culminated in critical recommendations concerning Rankine cycle power conversion, thermal control systems and advanced life support systems. Among others, key fluid physics mechanisms that have been identified are flow boiling and condensation. A key strategy is to develop mechanistic models for flow conditions that would ensure insensitivity to gravity level for these important heat transfer mechanisms. This would allow performing most of the required confirmatory experiments on Earth. Those topics and the minimum flow requirements constitute key objectives for the proposed study. Unfortunately, most of the two-phase heat transfer know-how amassed over nearly a century of research comes from experiments conducted in Earth's gravity. The proposed study aims to develop an experimentally validated, mechanistic model for the minimum flow criteria to ensure gravity independent flow boiling heat transfer and condensation. Parabolic flight experiments and preparations for ISS experiments will occur throughout the course of the proposed study. This project will be a joint effort between the Purdue University Boiling and Two-Phase Flow Laboratory (PU-BTPFL) and the NASA Glenn Research Center. Personnel from the two organizations combine extensive experience in research and development of both flow boiling and condensation systems, and in conducting microgravity experiments. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The proposed research will lay the foundation of development of advanced high power to weight ratio thermal management systems for NASA's future space vehicles and planetary bases. |
| **Task Progress:** | The second year of this project included a number of tasks addressing the effects of gravity on flow boiling and condensation, as well as interfacial instabilities. This report summarizes findings from two topics that were completed during the second year: condensation in vertical downflow, and condensation in upflow. These findings will be used during the third year of the project to assess the influence of body force on condensation. The report is presented in two parts.<br>Part 1 explores the interfacial and heat transfer characteristics of annular condensation of FC-72 in vertical downflow. Two separate condensation test modules are employed, one for high-speed video imaging of the film interface and the second for heat transfer measurements. Condensation in both test modules is achieved by rejecting the heat to a counterflow of cooling water. The heat transfer measurements are obtained along the inner wall of an 11.89-mm i.d. and 1,259.84-mm long stainless steel tube. For very low FC-72 flow rates, the film is observed to be both smooth and laminar. The film turns turbulent with a very wavy interface as the flow rate of FC-72 is increased, especially for exit film Reynolds numbers above 1,800. The heat transfer coefficient decreases axially because of a gradual thickening of the liquid film. However, the data show a downstream minimum before the heat transfer coefficient increases again towards the outlet as the film transitions to turbulent flow, enhanced by the more intense downstream waves. A control-volume-based model is proposed, which incorporates an eddy diffusivity profile for the liquid film that accounts for interfacial dampening of turbulence due to surface tension. The model shows good accuracy in predicting the average condensation heat transfer coefficient data, evidenced by a mean absolute error of 12.59%.<br><br>Part 2 explores upflow condensation in vertical tubes. This orientation is complicated by the relative magnitude of the opposing vapor shear and gravity. This study examines the different flow regimes for condensation of FC-72 in a vertical tube using both high-speed video imaging and detailed heat transfer measurements. Four regimes are identified, falling film, where the condensing film drains downwards by gravity opposite to low velocity vapor flow, oscillating film, corresponding to film flow oscillating between upwards and downwards, flooding, where film begins to be sheared upwards by the vapor core, and climbing film, where high vapor velocity causes the film to be sheared upwards. The four flow regimes are well segregated in a flow regime map based on dimensionless superficial velocities of the vapor and liquid. The condensation heat transfer coefficient is shown to decrease axially because of gradual thickening of the film, except for high mass velocities, where turbulence and intensified interfacial waviness cause downstream heat transfer enhancement. An annular flow model is constructed, which shows fair predictions for the climbing film regime. The predictive accuracy of the model is influenced by flow oscillations occurring downstream of the climbing film region and inability of the model to account for interfacial waves. |
| **Bibliography Type:** | Description: (Last Updated: 01/06/2026) |
| **Abstracts for Journals and Proceedings** | Mudawar I, Hasan MM. (additional authors per PI but not listed in program: Lee, H., Konishi, C., Kim, S.M., Hall, N., Nahra, H.) "Flow Boiling and Condensation Experiment (FBCE) for the International Space Station– Predictive Tools." 29th American Society for Gravitational and Space Research and 5th International Symposium for Physical Sciences in Space, Orlando, FL, November 3-8, 2013.<br>29th American Society for Gravitational and Space Research and 5th International Symposium for Physical Sciences in Space, Orlando, FL, November 3-8, 2013. Program and abstracts, p. 47.<br>https://asgsr.org/images/stories/pdf/Complete_Program_FINAL_2013_Meeting.pdf , Nov-2013 |
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| **Articles in Peer-reviewed Journals** | Lee H, Mudawar I, Hasan MM. "Flow condensation in horizontal tubes." International Journal of Heat and Mass Transfer. 2013 Nov;66:31-45. http://dx.doi.org/10.1016/j.ijheatmasstransfer.2013.06.044 , Nov-2013 |
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